

Laboratorio 6 - Repaso EBS y EC2

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Edit inbound rulesInfo

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rulesInfo

Security group rule ID

TypeInfo

ProtocolInfo

Port rangeInfo

SourceInfo

Description - optionalInfo

sgr-06de6f930922c2645

SSH

TCP

22

Cust...

Q

181.59.3.163/32

X

Linux-angel

Delete

Add rule

Cancel

Preview changes

Save rules

Launch an instanceInfo

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tagsInfo

Name

EC2-AngelGonzalez-EXAMEN2

Add additional tags

▼ Application and OS Images (Amazon Machine Image)Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Q

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debia

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-0f3caa1cf4417e51b (64-bit (x86), uefi-preferred) / ami-0bea3ccc607167c10 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260216.1 x86_64 HVM kernel-6.1

Architecture

64-bit (x...

Boot mode

uefi-preferred

AMI ID

ami-0f3caa1cf4417e51b

Publish Date

2026-02-16

Username

ec2-user

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

angelgonzalez-examen2

[Create new key pair](#)

▼ **Configure storage** [Info](#)

[Advanced](#)

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

[Click refresh to view backup information](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

[Edit](#)

► **Advanced details** [Info](#)

▼ Summary

Number of instances | Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0f3caa1cf4417e51b

Virtual server type (instance type)
t3.micro

Firewall (security group)
ec2-rds-AngelGonzalezPadillaExamen2

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Instance summary for i-01f557e1a4258ea31 (EC2-AngelGonzalez-EXAMEN2) Info

Updated less than a minute ago

Instance ID
i-01f557e1a4258ea31

IPv6 address
-

Hostname type
IP name: ip-172-31-14-253.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
100.55.7.75 [Public IP]

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
100.55.7.75 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-14-253.ec2.internal

Instance type
t3.micro

VPC ID
vpc-00f479057476a2db8 (VPC-Produccion)

Subnet ID
subnet-00227b47a5f26ec04

Instance ARN
arn:aws:ec2:us-east-1:654654478122:instance/i-01f557e1a4258ea31

Private IPv4 addresses
172.31.14.253

Public DNS
ec2-100-55-7-75.compute-1.amazonaws.com | open address

Elastic IP addresses
-

AWS Compute Optimizer finding
User: am:aws:iam::654654478122:user/students/anggonpad@gmail.com is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action
Retry

Auto Scaling Group name
-

Managed
false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details Info

AMI ID
ami-0f3caa1cf4417e51b

AMI name
al2023-ami-2023.10.20260216.1-kernel-6.1-x86_64

Stop protection

Monitoring
disabled

Allowed image
-

Launch time

Platform details
Linux/UNIX

Termination protection
Disabled

AMI location

Instance summary for i-01f557e1a4258ea31 (EC2-AngelGonzalez-EXAMEN2) Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID i-01f557e1a4258ea31	Public IPv4 address 100.55.7.75 open address	Private IPv4 addresses 172.31.14.253
IPv6 address -	Instance state Running	Public DNS ec2-100-55-7-75.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-14-253.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-14-253.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding User: arn:aws:iam::654654478122:user/students/anggonpad@gmail.com is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action Retry
Auto-assigned IP address 100.55.7.75 [Public IP]	VPC ID vpc-00f479057476a2db8 (VPC-Produccion)	Auto Scaling Group name -
IAM role -	Subnet ID subnet-00227b47a5f26ec04	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:654654478122:instance/i-01f557e1a4258ea31	
Operator -		

Creación volumen EBS

vol-07e06a510981fe5f0 (AngelGonzalez-EBSLAB6)

Last updated less than a minute ago

Actions

Delete

Modify

Details

Volume ID vol-07e06a510981fe5f0 (AngelGonzalez-EBSLAB6)	Size 1 GiB	Type gp3	Status check Okay
AWS Compute Optimizer finding This user is not authorized to call AWS Compute Optimizer. Retry	Volume state Available	IOPS 3000	Throughput 125
Fast snapshot restored No	Availability Zone use1-az1 (us-east-1a)	Created Thu Feb 26 2026 18:36:50 GMT-0500 (Colombia Standard Time)	Multi-Attach enabled No
Attached resources -	Outposts ARN -	Managed false	Operator -
Source			
Snapshot ID -	Source volume ID -		
Encryption			
Encryption Not encrypted	KMS key ID -	KMS key alias -	KMS key ARN -

Status checks

Monitoring

Tags

Successfully attached volume vol-07e06a510981fe5f0 to instance i-01f557e1a4258ea31.

Volumes (47) Info

Last updated less than a minute ago

Recycle Bin

Actions

Create volume

vol-07e06a510981fe5f0 (AngelGonzalez-EBSLAB6)

Last updated less than a minute ago

Actions

Delete

Modify

Details

Volume ID vol-07e06a510981fe5f0 (AngelGonzalez-EBSLAB6)	Size 1 GiB	Type gp3	Status check Insufficient data
AWS Compute Optimizer finding This user is not authorized to call AWS Compute Optimizer. Retry	Volume state In-use	IOPS 3000	Throughput 125
Fast snapshot restored No	Availability Zone use1-az1 (us-east-1a)	Created Thu Feb 26 2026 18:36:50 GMT-0500 (Colombia Standard Time)	Multi-Attach enabled No
Attached resources i-01f557e1a4258ea31 (EC2-AngelGonzalez-EXAMEN2) (EBS card index: 0): /dev/xvddb (attached)	Outposts ARN -	Managed false	Operator -

Source

Snapshot ID

Source volume ID

Al ingresar al EC2 verificamos

```
[ec2-user@ip-172-31-14-253 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO  TYPE MOUNTPOINTS
nvme0n1     259:0    0   8G  0  disk
├─nvme0n1p1 259:1    0   8G  0  part /
├─nvme0n1p127 259:2    0   1M  0  part
└─nvme0n1p128 259:3    0  10M  0  part /boot/efi
nvme1n1     259:4    0   1G  0  disk
[ec2-user@ip-172-31-14-253 ~]$
```

le damos formato

```
<value> is xxx (512 byte blocks).
[ec2-user@ip-172-31-14-253 ~]$ sudo mkfs -t xfs /dev/nvme1n1
meta-data=/dev/nvme1n1          isize=512    agcount=8, agsize=32768 blks
                               =               sectsz=512   attr=2, projid32bit=1
                               =               crc=1      finobt=1, sparse=1, rmapbt=0
                               =               reflink=1   bigtime=1 inobtcount=1 nrext64=0
                               =               exchange=0
data      =                     bsize=4096   blocks=262144, imaxpct=25
                               =               sunit=1     swidth=1 blks
naming    =version 2           bsize=4096   ascii-ci=0, ftype=1, parent=0
log       =internal log       bsize=4096   blocks=16384, version=2
                               =               sectsz=512   sunit=1 blks, lazy-count=1
realtime  =none                extsz=4096   blocks=0, rtextents=0
[ec2-user@ip-172-31-14-253 ~]$
```

Lo montamos en la EC2

```
[ec2-user@ip-172-31-14-253 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           459M   0  459M   0% /dev/shm
tmpfs           184M 440K  183M   1% /run
/dev/nvme0n1p1  8.0G 1.6G  6.4G  21% /
tmpfs           459M   0  459M   0% /tmp
/dev/nvme0n1p128 10M 1.3M  8.7M  13% /boot/efi
tmpfs           92M   0   92M   0% /run/user/1000
/dev/nvme1n1    960M  40M  921M   5% /mnt/logs
[ec2-user@ip-172-31-14-253 ~]$
```

Aumentamos el espacio

Modify volume info

Modify the type, size, and performance of an EBS volume.

Volume details

Volume ID

 vol-07e06a510981fe5f0 (AngelGonzalez-EBSLAB6)

Volume type

Info

General Purpose SSD (gp3) ▾

Size (GiB)

Info

3

Min: 1 GiB, Max: 65536 GiB.

IOPS

Info

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s)

Info

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Cancel

Modify

Ya enviada la solicitud esperamos

Requested volume modification for volume vol-07e06a510981fe5f0.

The volume is being modified.

Last updated less than a minute ago



 Recycle Bin

Actions ▾

Create volume

Volumes (1) Info

Saved filter sets

Choose filter set ▾

Q

Search

Volume ID = vol-07e06a510981fe5f0

X

Clear filters ▾

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Vol
☐	AngelGonzalez-EBSLAB6	vol-07e06a510981fe5f0	gp3	1 GiB	3000	125	-	-	2026/02/26 18:36 GMT-5	use1-az1 (us-east-1a)	

Ya se ejecutó la acción

Volumes (1) Info

Last updated less than a minute ago



 Recycle Bin

Actions ▾

Create volume

Volumes (1) Info

Saved filter sets

Choose filter set ▾

Q

Search

Volume ID = vol-07e06a510981fe5f0

X

Clear filters ▾

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Vol
☐	AngelGonzalez-EBSLAB6	vol-07e06a510981fe5f0	gp3	3 GiB	3000	125	-	-	2026/02/26 18:36 GMT-5	use1-az1 (us-east-1a)	

En la EC2 también se refleja

```
[ec2-user@ip-172-31-14-253 ~]$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1                             259:0    0   8G  0 disk
├─nvme0n1p1                         259:1    0   8G  0 part /
├─nvme0n1p127                       259:2    0   1M  0 part
└─nvme0n1p128                       259:3    0  10M  0 part /boot/efi
nvme1n1                             259:4    0   3G  0 disk /mnt/logs
[ec2-user@ip-172-31-14-253 ~]$
```

Se presenta el aumento al punto de montaje:

```
[ec2-user@ip-172-31-14-253 ~]$ sudo xfs_growfs /mnt/logs/
meta-data=/dev/nvme1n1             isize=512    agcount=8, agsize=32768 blks
      =                               sectsz=512   attr=2, projid32bit=1
      =                               crc=1      finobt=1, sparse=1, rmapbt=0
      =                               reflink=1   bigtime=1 inobtcount=1 nrext64=0
      =                               exchange=0
data      =                           bsize=4096   blocks=262144, imaxpct=25
      =                           sunit=1      swidth=1 blks
naming    =version 2                 bsize=4096   ascii-ci=0, ftype=1, parent=0
log        =internal log             bsize=4096   blocks=16384, version=2
      =                           sectsz=512   sunit=1 blks, lazy-count=1
realtime  =none                     extsz=4096   blocks=0, rtextents=0
data blocks changed from 262144 to 786432
[ec2-user@ip-172-31-14-253 ~]$
```

```
[ec2-user@ip-172-31-14-253 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           459M   0  459M   0% /dev/shm
tmpfs           184M 440K  183M   1% /run
/dev/nvme0n1p1  8.0G  1.6G  6.4G  21% /
tmpfs           459M   0  459M   0% /tmp
/dev/nvme0n1p128 10M  1.3M   8.7M  13% /boot/efi
tmpfs           92M   0   92M   0% /run/user/1000
/dev/nvme1n1    3.0G   55M  2.9G   2% /mnt/logs
[ec2-user@ip-172-31-14-253 ~]$
```

AUTOMATIZACIÓN

Creamos un script de bash, lo guardamos y cambiamos permisos para poderlo ejecutar

```
1 #!/bin/bash
2 echo "Se ejecuto en $(date)" >> $HOME/logs/cron_log.txt

test_cron.sh [+] 2:37
[ec2-user@ip-172-31-14-253 ~]$ sudo chmod +x test_cron.sh
[ec2-user@ip-172-31-14-253 ~]$ ls -l
total 4
drwxrwxr-x. 2 ec2-user ec2-user  6 Feb 27 00:24 logs
-rwxr-xr-x. 1 ec2-user ec2-user 68 Feb 27 00:21 test_cron.sh
[ec2-user@ip-172-31-14-253 ~]$

TERMINAL bash [-] 56% 9:32
-- TERMINAL --
[0] 0:nvim* "ip-172-31-14-253.ec2." 00:24 27-Feb-26
```


Verificamos ejecutando el script

```
[ec2-user@ip-172-31-14-253 ~]$ ./test_cron.sh
[ec2-user@ip-172-31-14-253 ~]$ tree
.
├── logs
│   └── cron_log.txt
└── test_cron.sh

1 directory, 2 files
[ec2-user@ip-172-31-14-253 ~]$ cat logs/cron_log.txt
Se ejecuto en Fri Feb 27 00:26:02 UTC 2026
[ec2-user@ip-172-31-14-253 ~]$
```

Instalamos y habilitamos el daemon de cron

```
[ec2-user@ip-172-31-14-253 ~]$ sudo systemctl enable crond
[ec2-user@ip-172-31-14-253 ~]$ sudo systemctl start crond
[ec2-user@ip-172-31-14-253 ~]$ sudo systemctl status crond
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-02-27 00:28:42 UTC; 8s ago
     Main PID: 30155 (crond)
       Tasks: 1 (limit: 1067)
      Memory: 1008.0K
         CPU: 9ms
      CGroup: /system.slice/crond.service
              └─30155 /usr/sbin/crond -n

Feb 27 00:28:42 ip-172-31-14-253.ec2.internal systemd[1]: Started crond.service - Command Scheduler.
Feb 27 00:28:42 ip-172-31-14-253.ec2.internal crond[30155]: (CRON) STARTUP (1.5.7)
Feb 27 00:28:42 ip-172-31-14-253.ec2.internal crond[30155]: (CRON) INFO (Syslog will be used instead of sendmail.)
Feb 27 00:28:42 ip-172-31-14-253.ec2.internal crond[30155]: (CRON) INFO (RANDOM_DELAY will be scaled with factor 65% if used.)
Feb 27 00:28:42 ip-172-31-14-253.ec2.internal crond[30155]: (CRON) INFO (running with inotify support)
[ec2-user@ip-172-31-14-253 ~]$
```

Creamos el cron

```
1 * * * * echo "Hola cron $(date)" >> $HOME/cron_log.txt
```

```
INSERT crontab.9DkcaT [+] utf-8 < Δ < ≡ crontab Top 1:57
— INSERT —
[0] 0:crontab* "ip-172-31-14-253.ec2." 00:32 27-Feb-26
```

```
[ec2-user@ip-172-31-14-253 ~]$ crontab -l
* * * * * echo "Hola cron $(date)" >> $HOME/cron_log.txt
[ec2-user@ip-172-31-14-253 ~]$ cat cron_log.txt
Hola cron Fri Feb 27 00:34:01 UTC 2026
Hola cron Fri Feb 27 00:35:01 UTC 2026
[ec2-user@ip-172-31-14-253 ~]$
```

[0] 0: bash*

"ip-172-31-14-253.ec2." 00:35 27-Feb-26